

## **Unnamed\_Unit**

### **Unit 4 - Variables**

#### **PDF Documents**

##### **PDF Document 1: 1734182293481-Chapter 3 - unit 4 - Variables.pdf**

This PDF document is included in the unit materials.

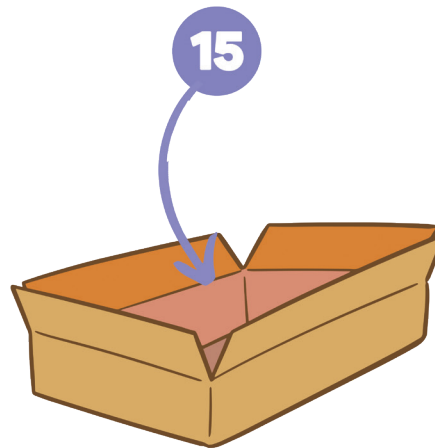
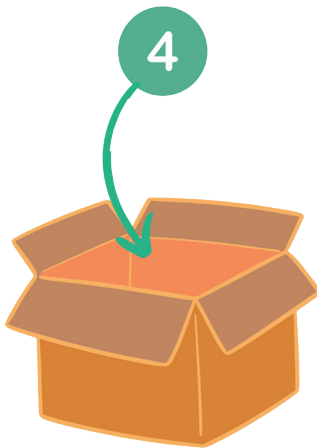


## UNIT 4 VARIABLES

### Variables

Variables are like labeled boxes in a computer where we keep information we need, like numbers or names. They help us remember and change things in our code, making it smarter.

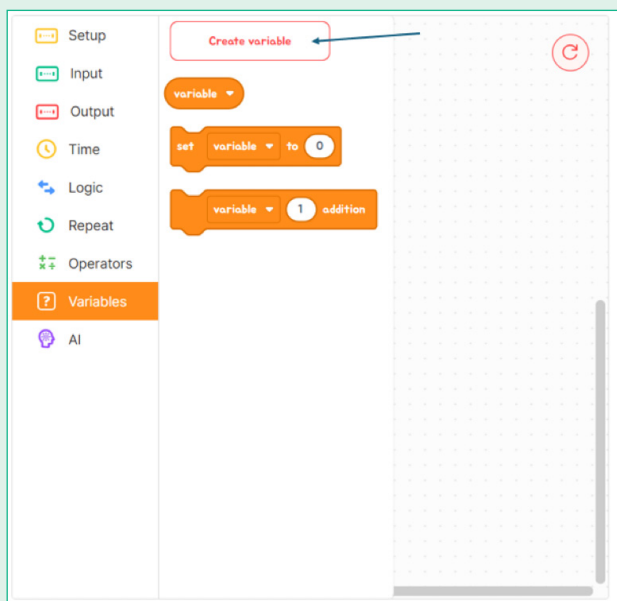
## ✓ VARIABLES



LET'S MAKE A VARIABLE IN CODE EDITOR:

#### Step1

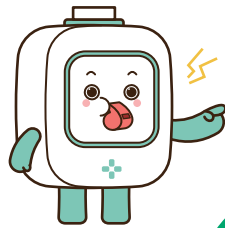
Go to the Variables section and click "Create Variable."





### Step2

Type the name for your variable (like "mode").

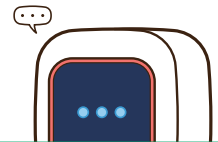


### Step3

After typing the name, click "Make" to create it.

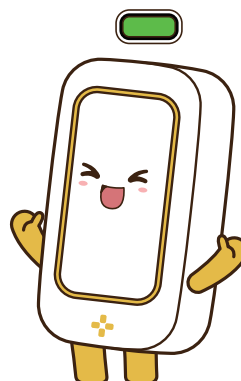
### Step4

You'll then see your new variable in the list.



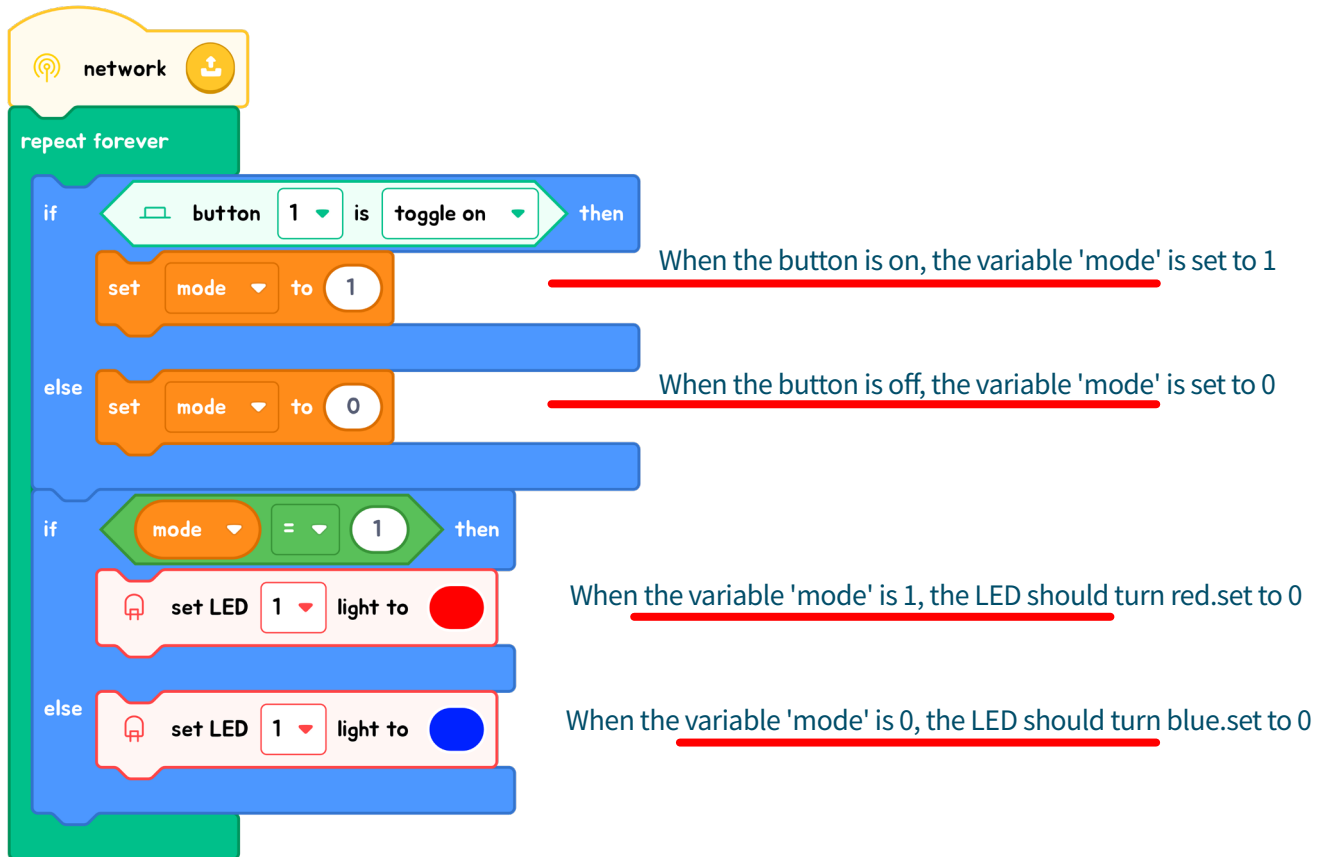
### Step5

Click on the area where variables are shown to see the one you made.



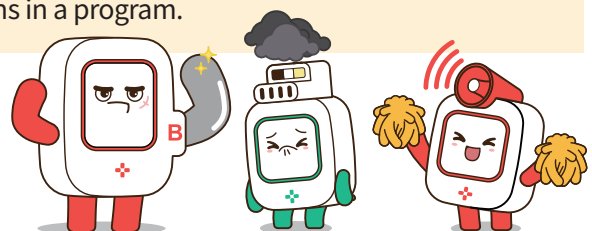
## EXPLANATION WITH CODE EXAMPLE

In our code, we're using a button and an LED light along with a variable named 'mode'. Here's what happens in our program:



## Code Insight

- 1. Variable 'mode':** Think of 'mode' as the program's memory of whether the button is on or off. 'Mode' can be either 1 (on) or 0 (off).
- 2. Checking the Button:** The first part of our code checks the button's state. If the button is on, 'mode' gets set to 1; if it's off, 'mode' goes to 0. This is how the program 'remembers' the button's state.
- 3. Action Based on 'mode':**
  - If 'mode' is 1: The program sees 'mode' is 1 and knows the button was on, so it turns the LED red.
  - If 'mode' is 0: The program finds 'mode' is 0, it makes the LED blue.
- 4. The Role of 'mode':** The variable 'mode' acts as a switch for the LED's color, changing based on the button's state, showing how variables can control actions in a program.

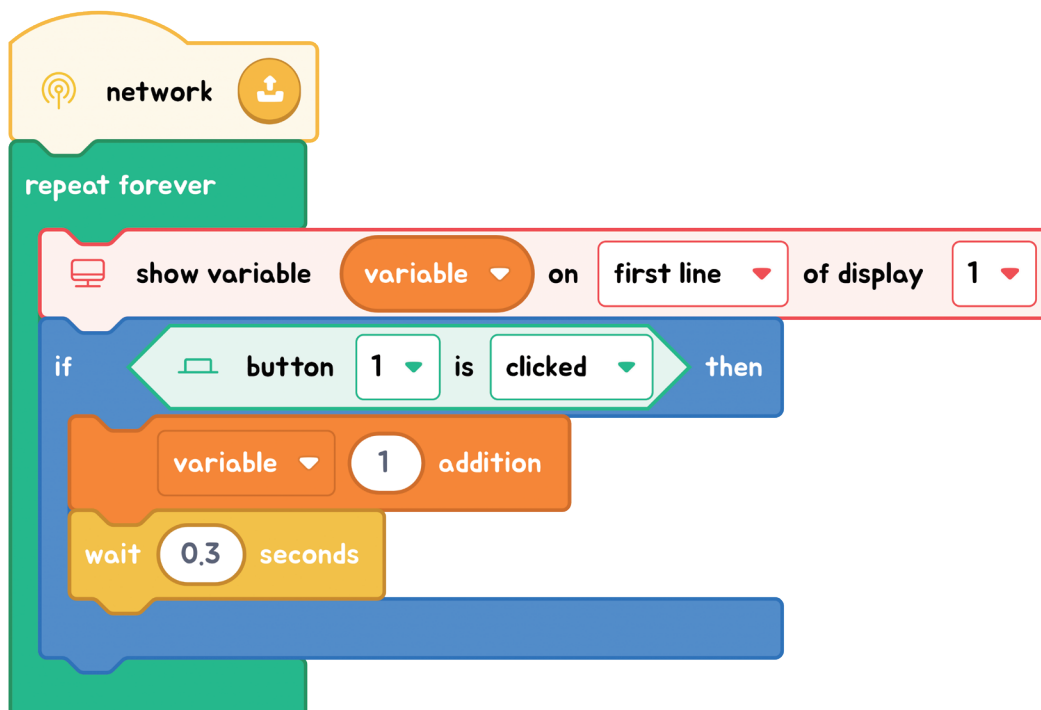




## Exercise

### Exercise: Click Counter challenge

Every time you click the button, we want to add one more to our click counter (that's our variable). Using a 'plus' block in block coding, we can keep track of how many times the button has been clicked. When we first make a variable, it starts at 0, unless we tell it to start at a different number.



#### Code Insight

1. **Starting Point:** When we begin, **our display screen is set to show a number(variable) called "clicks."** Think of "clicks" like your score in a video game, which everyone can see on the screen.
2. **Click to Add:** **Each time the button is pressed, we add 1 to our "clicks."** Because the display is always showing "clicks," everyone can see the score go up each time the button is clicked.
3. **Updating the Score:** Just like how a scoreboard updates every time a player scores in a game, **our display updates the "clicks" number every time the button is clicked, showing the new, higher score.**
4. **To decrease the number with each button click:** switch to using a subtraction operation in the code.

